

Stochastic-cascade parallel track: Hasegawa–Wakatani bridge

Living project note

1 September 2026

1 Purpose

This note records the exploratory stochastic-cascade parallel track to the canonical finite-dimensional nonmodal theory. The proposed chain is

field dynamics $\rightarrow$ zonal/nonzonal split $\rightarrow$ scale process $\rightarrow$ Markov test $\rightarrow$ Kramers–Moyal diagnostics.

It does not choose the final plasma convention for the main project and it does not constitute a converged turbulence calculation.

2 Benchmark equations

The modified Hasegawa–Wakatani (mHW) pilot uses

$$\partial_t \zeta + \{\phi, \zeta\} = \alpha(\tilde{\phi} - \tilde{n}) - D_\zeta \nabla^4 \zeta, \quad (1)$$

$$\partial_t n + \{\phi, n\} = \alpha(\tilde{\phi} - \tilde{n}) - \kappa \partial_y \phi - D_n \nabla^4 n, \quad (2)$$

with

$$\zeta = \nabla_\perp^2 \phi, \quad \tilde{f} = f - \langle f \rangle_y.$$

The Numata-style modification removes the resistive/adiabatic coupling from zonal modes $k_y = 0$. C2 also uses an original-HW control in which the same coupling is retained on zonal modes.

All C2 runs use a 32^2 periodic grid with $L_x = L_y = 40$, $\alpha = \kappa = 1$, $D_\zeta = D_n = 0.02$, a pseudo-spectral Poisson bracket, two-thirds dealiasing, RK4 with $\Delta t = 0.05$, and random initial perturbations of amplitude 10^{-3} .

3 C1: single-realization bridge pilot

The first pilot used seed 3 and the transient window $120 \leq t \leq 310$. It established that the full diagnostic chain can be implemented and gave an initial low/high-zonal-flow split. For the scale step $4\Delta x \rightarrow 2\Delta x$, fits of

$$D^{(1)}(u) \approx a_1 u + a_3 u^3, \quad D^{(2)}(u) \approx b_0 + b_2 u^2$$

gave

coefficient	low ZF	high ZF
a_1	-0.72497	-0.72289
a_3	0.00337	0.00207
b_0	0.34951	0.11664
b_2	0.19309	0.17742

The factor-of-three drop of b_0 was treated only as a hypothesis because ZF strength was strongly confounded with time.

4 C2: independent-seed falsification

C2 integrates four independent seeds, 0, 1, 2, 3, from $t = 0$ to 500 and analyzes only $380 \leq t \leq 490$. The mean zonal kinetic-energy fractions are

seed	mHW	HW
0	0.93080	0.05864
1	0.95111	0.04578
2	0.95755	0.04193
3	0.90398	0.04823

The mHW ZF fraction is near a late plateau, although the residual nonzonal energy still has seed-dependent slow trends. The window is therefore called late or quasi-stationary, not rigorously stationary.

4.1 Direction-resolved diffusion signal

Each mHW seed is split at its own median ZF fraction. Radial (x) and poloidal (y) increments are analyzed separately. Seed-level bootstrap intervals use 20,000 resamples of the four independent realization-level estimates.

For the $4\Delta x \rightarrow 2\Delta x$ step,

direction	mean $b_0^{\text{high}}/b_0^{\text{low}}$	95% interval
x	0.80188	[0.71009, 0.89366]
y	0.86447	[0.84857, 0.88448]

All four seeds have $b_0^{\text{high}} < b_0^{\text{low}}$ in both directions. The original C1 magnitude was therefore too large, but the sign of the diffusion modulation survives the independent-seed and anisotropy controls.

The corresponding radial a_1 change is consistent with zero; hence the dominant reproducible C2 signal remains diffusion-like rather than drift-like.

4.2 Markov-improvement hypothesis is falsified

A discrete Chapman–Kolmogorov test compares

$$P(X_2 | X_0) \quad \text{with} \quad P(X_2 | X_1)P(X_1 | X_0).$$

For the 8–4–2 triplet, the high-minus-low ZF change in occupancy-weighted total-variation error is

$$\Delta_{\text{CK}}^x = -0.00306 \quad [-0.00745, 0.00452],$$

while

$$\Delta_{\text{CK}}^y = +0.01341 \quad [0.00366, 0.02119].$$

Thus stronger ZF does not generally improve Markov closure. The radial change is unresolved at this seed count and the poloidal error becomes worse. Markov closure must remain scale- and direction-dependent and is an independent gate.

4.3 Original-HW control

Each model, seed and direction is normalized by its own $r = 8\Delta x$ standard deviation so that the comparison is not merely a comparison of raw fluctuation amplitudes. Even after this self-normalization, mHW has lower finite-step diffusion intercept than HW:

step	x : mHW/HW	95% interval	y : mHW/HW	95% interval
$8 \rightarrow 7$	0.70686	[0.67239, 0.73490]	0.52698	[0.48195, 0.57200]
$8 \rightarrow 4$	0.73844	[0.64518, 0.83361]	0.61841	[0.57258, 0.68577]

The conservative interpretation is therefore

strong zonal organization is associated with reduced finite-step stochastic spreading of the nonzonal scale pro

This remains an association statement: changing from HW to mHW changes the equations, not only the instantaneous ZF amplitude.

5 Small-step and Pawula gate

At fixed current separation $r = 8\Delta x$, C2 evaluates next separations

$$r' = 7, 6, 5, 4 \Delta x, \quad \Delta s = \ln(8/r').$$

The smallest available scale step is $\ln(8/7) \simeq 0.1335$.

The mean fitted b_0 values do not form a scale-step-independent plateau:

case	$8 \rightarrow 7$	$8 \rightarrow 6$	$8 \rightarrow 5$	$8 \rightarrow 4$
mHW x	0.52627	0.66145	0.55804	0.41018
mHW y	0.51473	0.65760	0.61463	0.45771
HW x	0.74600	0.92691	0.77099	0.55781
HW y	0.98276	1.21100	1.01931	0.74598

For the finite-step fourth coefficient

$$D_{\Delta s}^{(4)}(u) = \frac{\langle (\Delta u)^4 | u \rangle}{24 \Delta s},$$

a simple linear extrapolation of its constant fit coefficient gives nonzero $\Delta s \rightarrow 0$ intercepts:

case	extrapolated c_0	95% interval
mHW x	0.03186	[0.02945, 0.03426]
mHW y	0.02196	[0.01266, 0.03125]
HW x	0.05968	[0.05430, 0.06444]
HW y	0.09816	[0.08597, 0.11034]

The extrapolation itself is only a resolution diagnostic. The appropriate conclusion is that the present 32^2 data do not justify a Pawula/Fokker–Planck truncation.

6 C3: resolution and scale-coordinate gate

C3 tests whether the failed small-step gate is mainly a resolution problem or a coordinate problem. The mHW benchmark is repeated at 64^2 . To keep the physical separations fixed relative to C2, the increment sequence becomes

$$16 \rightarrow 14, 12, 10, 8$$

grid points, corresponding to $r = 10 \rightarrow 8.75, 7.5, 6.25, 5$.

Seed 1 undergoes a strong late reorganization and is excluded from the plateau-coordinate comparison; seeds 0, 2 and 3 have late ZF fractions near a plateau. A dedicated seed-1 control with half the time step, $\Delta t = 0.025$, reproduces the $380 \leq t \leq 490$ statistics to approximately four significant digits, so the analysis-window reorganization is not a visible RK4-step artifact.

6.1 Resolution-only test

The mean fitted b_0 values over plateau-like seeds 0, 2 and 3 are

case	10 $\rightarrow$ 8.75	10 $\rightarrow$ 7.5	10 $\rightarrow$ 6.25	10 $\rightarrow$ 5
64^2 mHW x	0.2580	0.3498	0.3306	0.2797
64^2 mHW y	0.1429	0.2187	0.2180	0.1660

No small-step plateau appears. The mean within-seed coefficient of variation across these steps is about 0.164 in x and 0.210 in y .

The fitted fourth-order extrapolation becomes smaller on average,

$$\bar{c}_{0,x} \simeq 0.00746, \quad \bar{c}_{0,y} \simeq 0.00656,$$

versus 0.03186 and 0.02196 at 32^2 , but the individual fits scatter strongly. Therefore resolution improves the Pawula tendency without establishing convergence.

6.2 Smooth Gaussian coarse-graining

Define a nested filtered field

$$X_\ell = \mathcal{F}^{-1} \left[\exp \left(-\frac{1}{2} \left(\frac{k\ell}{2\pi} \right)^2 \right) \hat{\phi}_{NZ} \right].$$

For $\ell = 10, 7.5, 5$, the mean Chapman–Kolmogorov TV error is 0.0389, compared with 0.1042 and 0.1041 for the radial and poloidal increment coordinates at comparable physical scales. Thus the smooth coordinate appears substantially more Markovian.

However its finite-step moments scale as

$$D_{\Delta s}^{(2)}(0) \propto \Delta s^{p_2}, \quad p_2 \in [0.78, 0.83],$$

and

$$D_{\Delta s}^{(4)}(0) \propto \Delta s^{p_4}, \quad p_4 \in [2.68, 2.86].$$

This is close to the differentiable-scale expectation $\Delta X = O(\Delta s)$, for which $D^{(2)} = O(\Delta s)$ and $D^{(4)} = O(\Delta s^3)$. Hence the improved closure is bought by a degenerate diffusion limit,

$$\lim_{\Delta s \rightarrow 0} D_{\Delta s}^{(2)} = 0.$$

The Gaussian coordinate is therefore rejected as a nontrivial Fokker–Planck cascade coordinate.

6.3 Sharp spectral cumulative coordinate

A non-smooth control,

$$X_K = \sum_{|k| \leq K, k_y \neq 0} \phi_k e^{ik \cdot x},$$

with $K = 0.6, 0.7, 0.8, 0.9, 1.0$ gives a mean CK error of about 0.0896, only modestly below the increment value. Its central $D^{(2)}$ is again strongly step dependent and no nonzero plateau is observed.

6.4 C3 methodological conclusion

C3 shows that “more Markovian” is not by itself a sufficient coordinate-selection criterion. A useful scale representation must satisfy both

$$\text{closure} \quad \text{and} \quad \text{non-degenerate generator behavior.}$$

A smooth coordinate may reduce conditional memory while trivializing the stochastic generator. Future mode/coordinate objectives therefore need a Markov-closure term together with a generator regularity or non-degeneracy constraint.

7 Current gate decision

- **Passed and retained:** the finite-scale ZF-conditioned transition-kernel viewpoint remains meaningful.
- **Falsified:** stronger ZF does not generally improve Markov closure.
- **C3a failed:** 64^2 alone does not produce a step-independent $D^{(2)}$ plateau.
- **C3b Gaussian rejected:** better CK closure is accompanied by $D^{(2)} \rightarrow 0$, i.e. a differentiable and degenerate scale flow.
- **C3b sharp spectral failed:** the cumulative spectral coordinate remains strongly step dependent.
- **GENE transfer of a Fokker–Planck claim remains blocked.**

The next branch should compare scale-normalized orthogonal/wavelet variables with an explicitly non-Markov generalized-Langevin or Mori–Zwanzig description. This is now a model-selection problem, not merely a resolution problem.

8 References

R. Numata, R. Ball, and R. L. Dewar, *Nonlinear Simulation of Drift Wave Turbulence*, arXiv:physics/0703274 (2007).

A. Hakim, *Solving (Modified) Hasegawa–Wakatani equations*, Simulation Journal JE17 (2026).